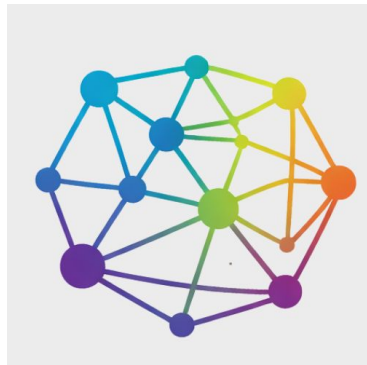




Sentinel-2 Surface Reflectance

Exploratory Data Analysis and Descriptive Analytics

Detailed project explanation based on the implemented Google Colab notebook

AI-powered carbon additionality, plantation monitoring and forest-change detection



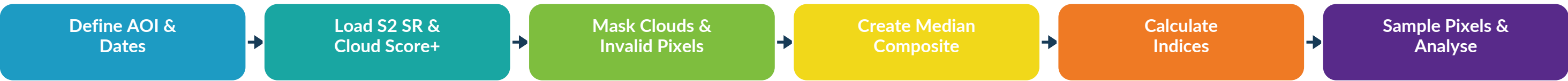
Notebook execution status: completed against Google Earth Engine with 236 filtered scenes and 5,000 sampled pixels.

Analysis window: 1 January 2025 to 31 December 2025 | AOI: 390.50 km² | Scale: 20 m



1. Executive Project Overview

The notebook is an end-to-end exploratory workflow for understanding Sentinel-2 Level-2A Surface Reflectance data before building a production carbon-monitoring model. It combines scene-level quality analysis, cloud masking, spectral-index engineering, pixel sampling, descriptive analytics, time-series analysis, land-cover interpretation and export preparation.



Component	Implemented approach	Purpose in carbon monitoring
Data source	COPERNICUS/S2_SR_HARMONIZED linked with Cloud Score+	Consistent surface-reflectance observations with pixel-level quality information
Study area	Demonstration polygon in central India; 390.50 km ²	Spatial boundary for plantation and forest-condition analysis
Observation period	Calendar year 2025	Captures pre-monsoon, monsoon and post-monsoon behaviour
Analytical scale	20 m	Harmonises bands and balances detail with processing cost
Sample	5,000 pixels from a median composite	Supports fast EDA without downloading the complete raster

EDA establishes whether the imagery is usable and what patterns exist. It does not by itself prove carbon additionality or certify claimable carbon.

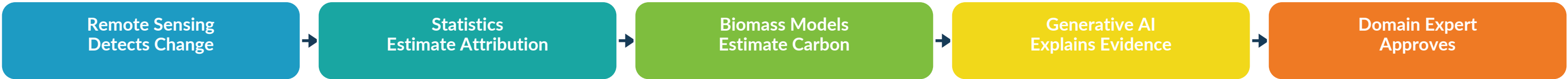


2. Business Problem and Project Context

Forestry carbon projects cannot claim all carbon visible after implementation. The project must distinguish pre-existing vegetation from post-project growth, compare the project area with a similar untreated reference region, and detect temporary loss events such as cutting followed by regrowth.

Business question	How Sentinel-2 EDA contributes	What remains required
Was vegetation already present?	Historical composites and NDVI-related distributions reveal baseline vegetation patterns.	Pre-project imagery, field inventory and plantation dates.
Did vegetation increase after intervention?	Monthly NDVI, EVI, NDMI and NBR reveal greening, moisture and disturbance patterns.	Multi-year pre/post comparison and a reference area.
Was there cutting or disturbance?	Abrupt NBR, NDMI and NDVI changes can flag candidate events.	Higher-frequency time series, Sentinel-1 and field verification.
How much carbon is additional?	EDA identifies useful predictors and data-quality limits.	Biomass calibration, causal comparison, uncertainty and methodology rules.

Correct analytical positioning





3. Dataset, Configuration and Area of Interest

Parameter	Implemented value	Meaning
Date range	2025-01-01 to 2025-12-31	A complete annual cycle for seasonal EDA
Maximum scene cloud	80%	Broad scene filter; pixel-level masking performs the stricter control
Cloud Score+ threshold	0.60	Pixels below the clear-quality threshold are masked
Analysis scale	20 metres	Compatible with red-edge and SWIR analysis
Pixel sample	5,000	Manageable client-side EDA sample
AOI area	390.50 km ²	Demonstration project boundary in central India

Core collection and preprocessing logic

```
s2_raw = (ee.ImageCollection('COPERNICUS/S2_SR_HARMONIZED')
    .filterBounds(aoi)
    .filterDate(start_date, end_date)
    .filter(ee.Filter.lte('CLOUDY_PIXEL_PERCENTAGE', 80)))

s2_linked = s2_raw.linkCollection(cloud_score, ['cs', 'cs_cdf'])
scaled_reflectance = image.select(spectral_bands).multiply(0.0001)
```

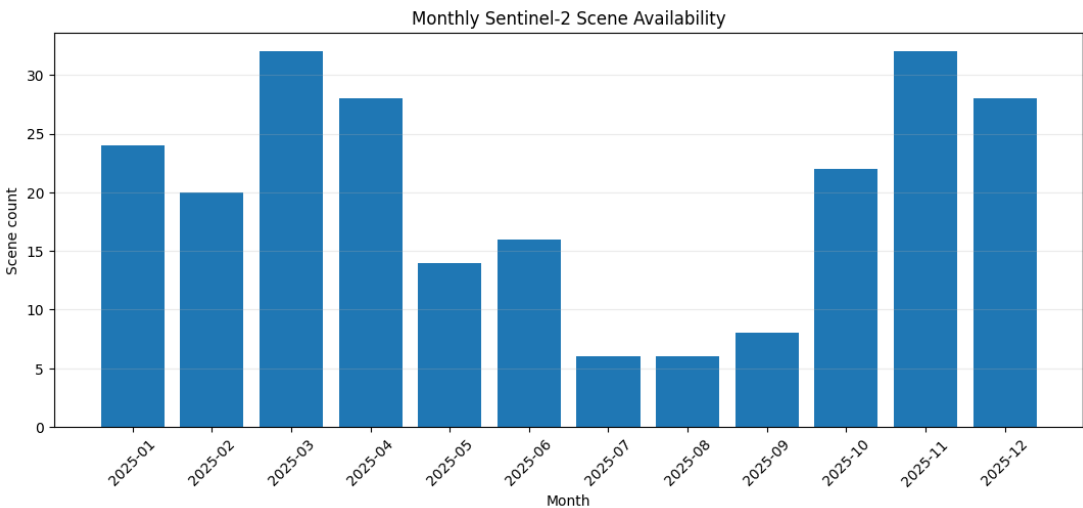
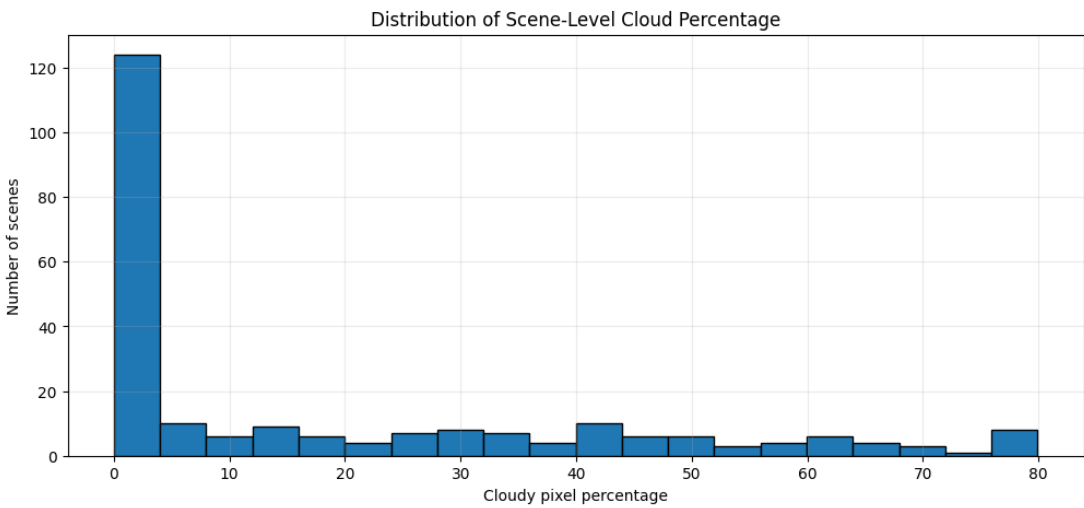
The notebook correctly converts integer band values to reflectance using a 0.0001 scale factor. It also links Cloud Score+ through the shared `system:index`, allowing per-pixel clear-quality masking.



4. Scene-Level Exploratory Data Analysis

After spatial, date and metadata filtering, the notebook found 236 Sentinel-2 scenes. All inspected metadata fields were complete, so there were no missing values in cloud percentage, solar angle, incidence angle, tile, processing baseline, spacecraft or timestamp.

Metric	Mean	Median	95th percentile	Maximum
Cloudy pixel percentage	18.03%	2.07%	68.39%	79.93%
Solar zenith angle	33.19°	34.08°	48.04°	48.87°
B8 incidence zenith angle	6.68°	6.75°	7.99°	8.06°



Key interpretation: annual median cloudiness was low, but July and August were extremely cloudy. Monthly median cloud percentage reached 70.91% in July and 76.20% in August. Optical monitoring during monsoon therefore requires strong pixel masking and should be supplemented with Sentinel-1 radar.



5. Seasonal Data Availability and Cloud Risk

Month	Scenes	Median cloud %	Analytical implication
Jan	24	0.83	Excellent optical coverage
Feb	20	0.01	Excellent optical coverage
Mar	32	0.00	Excellent optical coverage
Apr	28	0.20	Excellent optical coverage
May	14	40.51	High-risk; add radar / composites
Jun	16	25.86	Usable with masking
Jul	6	70.91	High-risk; add radar / composites
Aug	6	76.20	High-risk; add radar / composites
Sep	8	35.22	Usable with masking
Oct	22	36.44	Usable with masking
Nov	32	1.49	Excellent optical coverage
Dec	28	0.69	Excellent optical coverage

The temporal pattern is not a model result; it is a data-availability constraint that directly affects monitoring confidence.

Recommended production rule

Create monthly or seasonal clear-pixel composites, record observation counts per pixel, and flag periods where valid optical coverage falls below an agreed threshold. Use Sentinel-1 to maintain continuity during monsoon and persistent cloud periods.



6. Cloud and Invalid-Pixel Masking

The notebook combines Cloud Score+ with the Sentinel-2 Scene Classification Layer. This is stronger than relying only on scene-level cloud percentage.

Control	Implementation	Reason
Cloud Score+	Keep pixels where $cs_cdf \geq 0.60$	Removes low-confidence cloudy or hazy observations at pixel level
SCL no data	Exclude class 0	Avoid invalid measurements
Saturated/defective	Exclude class 1	Prevents sensor-quality artefacts
Cloud shadow	Exclude class 3	Shadows can mimic low vegetation or disturbance
Clouds/cirrus	Exclude classes 8, 9 and 10	Reduces false changes in indices
Snow/ice	Exclude class 11	Avoids non-vegetation spectral confusion

```
clear_mask = image.select('cs_cdf').gte(0.60)
valid_scl = (scl.neq(0).And(scl.neq(1)).And(scl.neq(3))
            .And(scl.neq(8)).And(scl.neq(9))
            .And(scl.neq(10)).And(scl.neq(11)))

return image.updateMask(clear_mask).updateMask(valid_scl)
```

Review note: a fixed 0.60 threshold is appropriate for an exploratory workflow, but production calibration should compare alternative thresholds against manually reviewed clear/cloud labels for the specific region and season.



7. Spectral Feature Engineering

The notebook calculates seven indices that describe greenness, canopy vigour, moisture, disturbance, water and exposed soil. Together they provide a more complete forest-condition signal than NDVI alone.

Index	Bands / concept	Project interpretation
NDVI	$(B8 - B4) / (B8 + B4)$	Greenness and canopy development
EVI	NIR, red and blue with atmospheric/soil adjustment	Vegetation vigour where NDVI may saturate
NDMI	$(B8 - B11) / (B8 + B11)$	Canopy moisture and drought stress
NBR	$(B8 - B12) / (B8 + B12)$	Burn, canopy loss and disturbance screening
NDWI	$(B3 - B8) / (B3 + B8)$	Water and wet-surface separation
SAVI	NDVI-style ratio with soil correction	Sparse plantation and exposed-soil areas
BSI	SWIR + red versus NIR + blue	Bare soil, clearing and land preparation

In carbon monitoring, a single index is evidence of spectral change - not proof of tree age, biomass, additionality or credit eligibility.



8. Pixel Sample Quality and Descriptive Analytics

The median composite was sampled at 20 m to create a 5,000-row analytical table containing ten reflectance bands and seven indices. The sample contained no missing values and no values outside broad physical quality ranges. Nine duplicate rows were detected, representing only 0.18% of the sample.

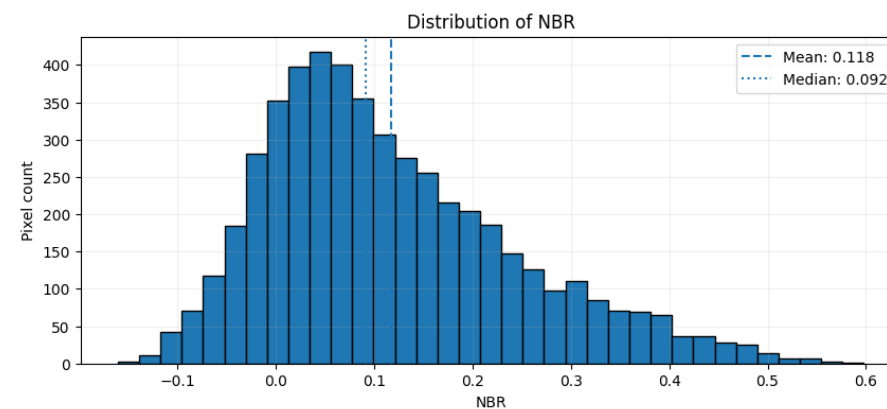
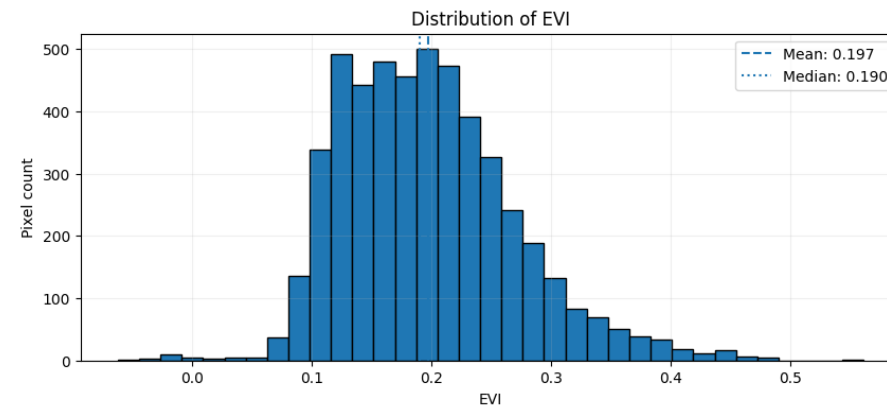
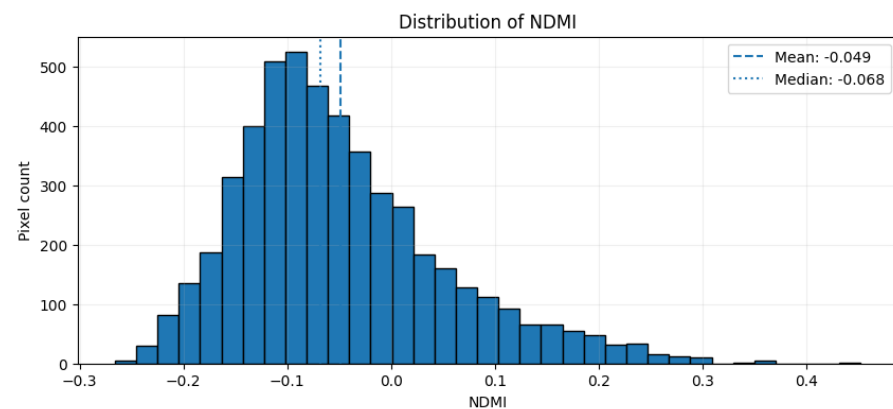
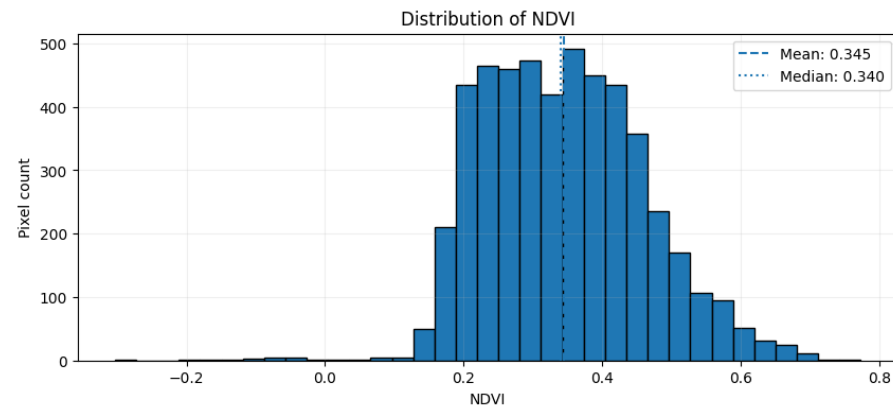
Check	Result	Assessment
Sample size	5,000 pixels	Sufficient for exploratory distributions and correlation analysis
Variables	17	10 spectral bands + 7 indices
Missing values	0	Complete analytical sample after masking
Duplicate rows	9	Negligible; may represent repeated composite values
Reflectance outside [0, 1.5]	0	No broad reflectance anomalies
Indices outside [-1, 1]	0	Index calculations remained within expected range

Variable	Mean	Median	Std. dev.	Min	Max
NDVI	0.3448	0.3399	0.1154	-0.3034	0.7731
EVI	0.1973	0.1904	0.0722	-0.0617	0.5610
NDMI	-0.0491	-0.0677	0.1002	-0.2661	0.4525
NBR	0.1176	0.0917	0.1282	-0.1605	0.5979
BSI	0.1069	0.1250	0.0895	-0.2599	0.2836



9. Distribution and Outlier Interpretation

The notebook generates separate histograms for all seven indices and a combined boxplot. The distributions help identify land-cover mixtures, skewness, long tails and possible anomalies.

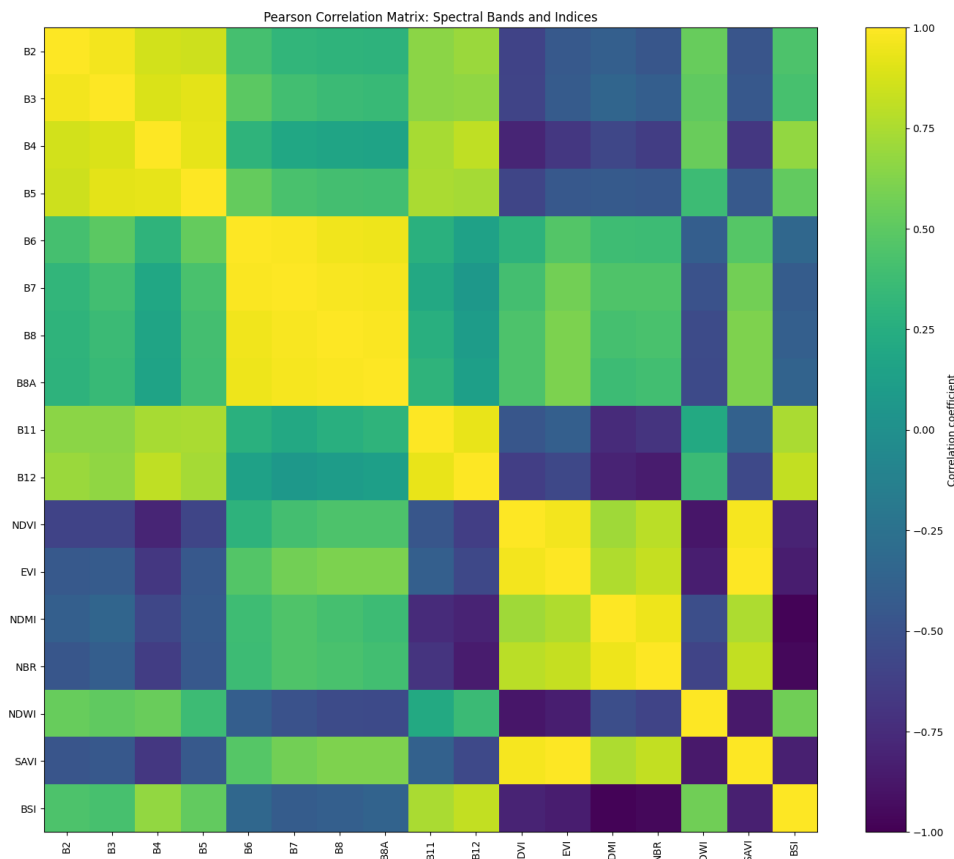


Interpretation: NDVI and EVI are centred in the sparse-to-moderate vegetation range. NDMI is slightly negative on average, indicating comparatively dry canopy/soil conditions in the annual median composite. NBR is mostly positive, while negative values represent a smaller group of potentially disturbed, non-vegetated or moisture-poor pixels.

Outlier caution: in remote sensing, an IQR outlier is not automatically an erroneous observation. It may represent water, settlement, exposed soil, dense canopy or a genuine disturbance. Spatial review is required before removal.



10. Correlation Analysis and Feature Relationships



Variable	Correlation with NDVI	Meaning
SAVI	0.9729	Very similar vegetation signal with soil adjustment
EVI	0.9676	Strong agreement in canopy-vigour measurement
NDWI	-0.8784	Higher greenness generally corresponds to more negative green-NIR water contrast
BSI	-0.8046	Vegetation increases as exposed-soil signal decreases
NBR	0.7955	Healthy canopy and disturbance signal are strongly related
NDMI	0.7158	Greener areas also tend to retain more moisture

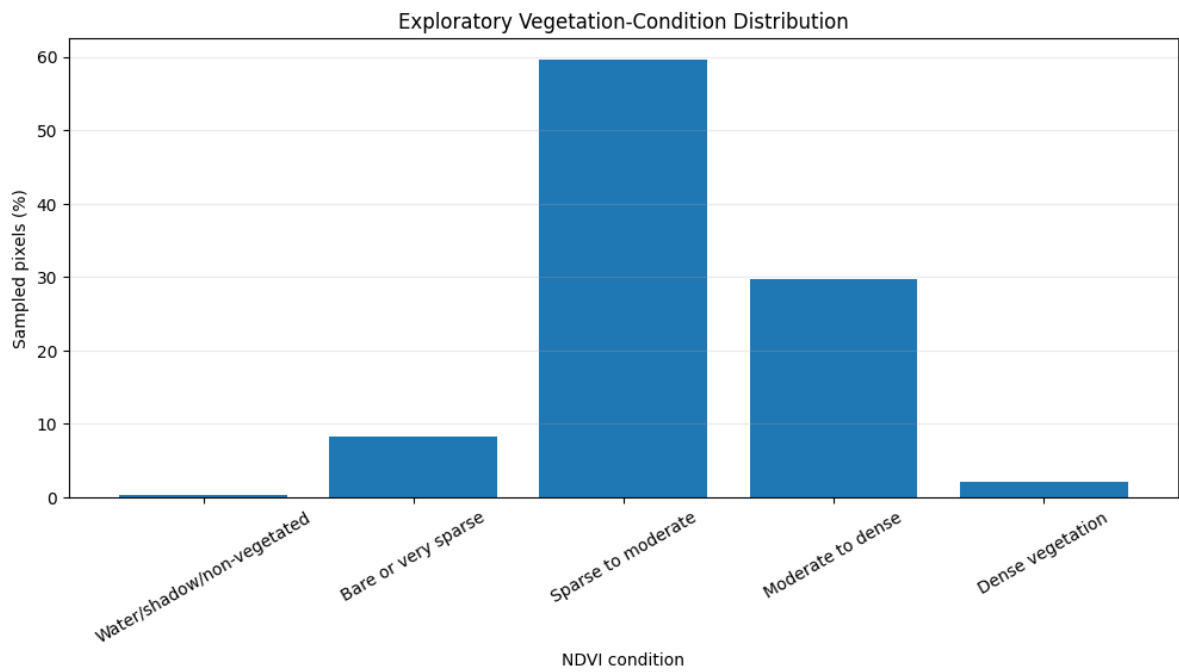
High correlation means some engineered features are redundant. Feature selection or regularisation should be considered before predictive modelling.



11. Vegetation Condition and Area Composition

The notebook applies exploratory NDVI thresholds to describe the annual median composite. The thresholds are not universal ecological classes and must be calibrated for local species, season, soil and plantation age.

Condition	Sample pixels	Pixel share	Estimated area	Area share
Water/shadow/non-vegetated	17	0.34%	134.37 ha	0.35%
Bare or very sparse	411	8.22%	2,963.00 ha	7.61%
Sparse to moderate	2,978	59.56%	23,378.09 ha	60.02%
Moderate to dense	1,489	29.78%	11,703.68 ha	30.05%
Dense vegetation	105	2.10%	769.03 ha	1.97%

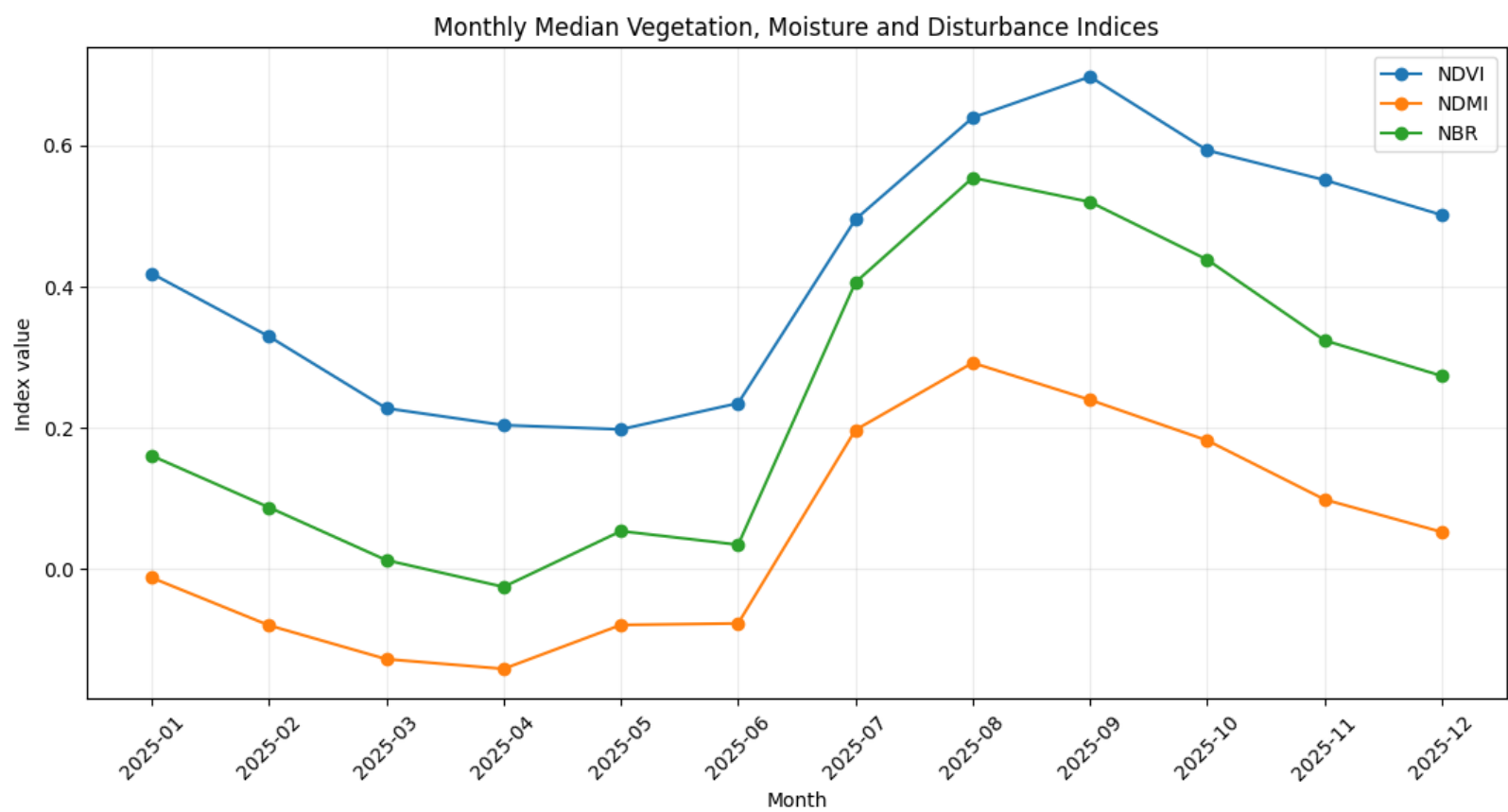


Main finding

The landscape is dominated by sparse-to-moderate vegetation. Dense vegetation accounts for only about 2% of sampled pixels and estimated area. This supports the notebook’s automated interpretation, but the annual median composite cannot distinguish plantation age or project causality.



12. Monthly Vegetation, Moisture and Disturbance Trends

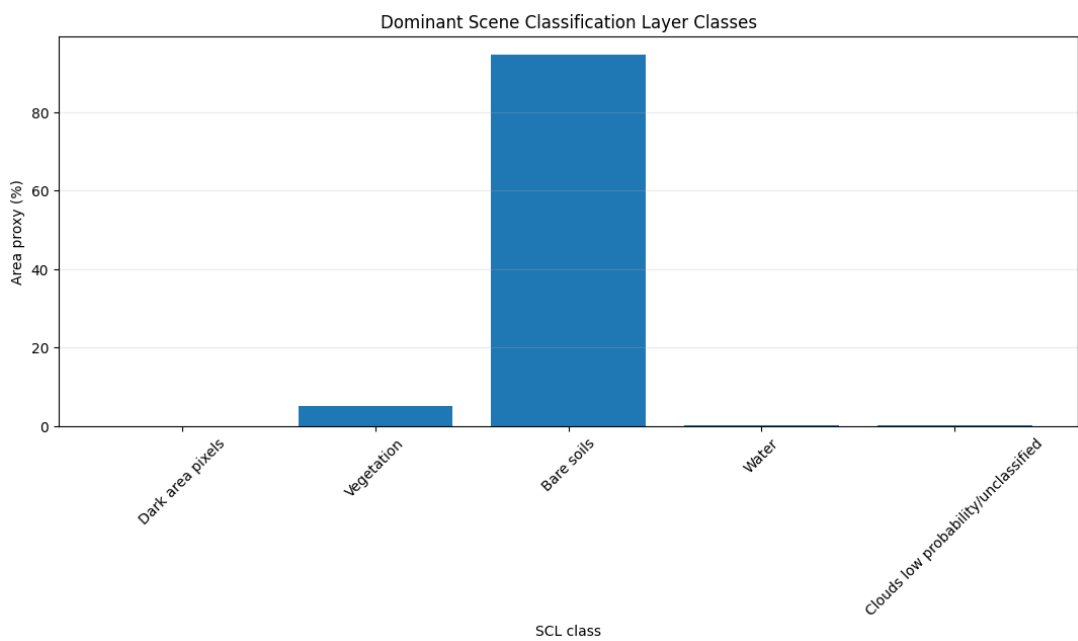


Period	Observed pattern	
January-April	NDVI falls from 0.418 to 0.204; NDMI becomes more negative	Dry-
May-June	NDVI remains low near 0.20-0.23	Pre-r requ
July-September	NDVI rises from 0.495 to 0.697; NDMI and NBR rise strongly	Mon imag
October-December	NDVI declines gradually from 0.593 to 0.501	Post

The strong seasonal cycle demonstrates why a simple before-versus-after image comparison can create false carbon-growth conclusions.



13. Scene Classification Layer Findings and an Important Limitation



Mode SCL class	Share
Bare soils	94.63%
Vegetation	5.08%
Water	0.22%
Low-probability cloud/unclassified	0.05%
Dark area pixels	0.02%

Why this conflicts with NDVI: the SCL mode composite labels 94.63% of the area as bare soil, while NDVI analysis classifies around 90% as sparse-to-dense vegetation. This does not necessarily mean the NDVI result is wrong. The mode of discrete SCL classes across many dates can be dominated by frequent dry-season classifications, tile-edge behaviour or classification uncertainty.

Required improvement: do not treat the annual mode SCL class as final land-cover truth. Analyse SCL by month, calculate valid observation counts, compare with Dynamic World or ESA WorldCover, and manually inspect representative pixels.



14. Consolidated Findings from the Implemented Notebook

Finding	Observed result	Business interpretation
Filtered scenes	236	Strong annual archive, but uneven monthly usability
Median scene cloud	2.07%	Generally manageable after masking
Sampled pixels	5,000	Good exploratory sample; not full-pixel certification evidence
Mean NDVI	0.3448	Landscape is mainly sparse-to-moderate vegetation
Mean NDMI	-0.0491	Annual composite indicates relatively dry conditions
Mean NBR	0.1176	Mostly positive disturbance/vegetation signal with local low values
Dense vegetation	2.10%	Dense canopy is a small proportion of the analysed area
Additionality	Not established	Requires baseline, reference area and causal comparison

Overall conclusion

The notebook successfully demonstrates data ingestion, quality control and descriptive remote-sensing analytics. It is a sound EDA foundation, but it remains an exploratory single-area, single-year analysis rather than a carbon additionality or biomass-estimation system.



15. Outputs, Export Workflow and Observed Error

The notebook successfully writes analytical tables to CSV. It also attempts to download the composite and indices as one GeoTIFF. The direct download fails because the requested file is approximately 144.5 MB, while the direct request limit is about 50.3 MB.

Output	Status	Recommended action
Scene metadata CSV	Successful	Use for coverage and cloud analysis
Pixel sample CSV	Successful	Use for local statistics and modelling experiments
Descriptive/correlation summaries	Successful	Use in reports and model-design decisions
Direct GeoTIFF download	Failed: request too large	Use Drive/Cloud Storage export, reduce region, increase scale or export bands separately
Google Drive batch export	Prepared but not started	Uncomment and start the batch task for large AOIs

```
task = ee.batch.Export.image.toDrive(  
    image=composite_indices.select(analysis_bands),  
    description='Sentinel2_SR_EDA_Composite',  
    folder='EarthEngineExports',  
    region=aoi,  
    scale=20,  
    maxPixels=1e13,  
    fileFormat='GeoTIFF')  
task.start()
```

For production, exports should be tiled and versioned. Each export must store the AOI, date window, masking threshold, index definitions, data collection IDs and code version to preserve auditability.



16. Technical Strengths, Constraints and Improvements

Area	Strength in notebook	Improvement for production
Authentication	Project-based Earth Engine initialization is implemented	Use service accounts or controlled credentials for scheduled workflows
Quality control	Cloud Score+ and SCL masking are combined	Calibrate threshold and quantify clear-observation counts
Feature engineering	Seven meaningful indices are calculated	Add red-edge indices, texture, Sentinel-1 and topographic features
EDA design	Metadata, quality, distributions, outliers and correlations are covered	Add spatial stratification and class-balanced sampling
Time series	Monthly trends reveal seasonality	Use multi-year baselines and change-point detection
Land-cover interpretation	NDVI and SCL summaries are provided	Resolve SCL contradiction using monthly analysis and external land-cover labels
Carbon relevance	Findings are correctly described as exploratory	Add reference area, field plots, GEDI biomass and methodology deductions

Major analytical constraints

The analysis uses a demonstration AOI, one calendar year, a median composite and threshold-based vegetation classes. It does not include pre-project years, a reference region, field measurements, species information, biomass labels, causal attribution, uncertainty intervals or carbon-accounting rules.



17. Recommended Production Architecture



Layer	Responsibility	Candidate technologies
Data ingestion	Acquire and catalogue imagery and project records	Earth Engine, STAC, Python, geemap
Geospatial processing	Mask, align, composite and calculate spatial features	Rasterio, GDAL, Xarray, GeoPandas
Change analytics	Detect growth, loss, cutting and regrowth	Time-series statistics, change-point models, PyTorch
Biomass modelling	Convert remote-sensing features into biomass estimates	GEDI, field plots, XGBoost, uncertainty models
Additionality	Compare project and matched reference changes	Difference-in-differences, matching, causal inference
Reporting	Provide maps, confidence, evidence and explanations	FastAPI, PostGIS, React, RAG/LLM



18. Interview-Ready Project Explanation

30-second explanation

I implemented an exploratory analysis of Sentinel-2 Level-2A Surface Reflectance for a carbon-monitoring use case. The notebook loads 2025 imagery from Google Earth Engine, links Cloud Score+ for pixel-level quality masking, removes cloud and invalid SCL classes, scales reflectance, creates a median composite and calculates NDVI, EVI, NDMI, NBR, NDWI, SAVI and BSI. I then analyse 236 scenes and a 5,000-pixel sample using descriptive statistics, distributions, outlier checks, correlations, monthly trends and vegetation-condition classes. The analysis found strong seasonality and a landscape dominated by sparse-to-moderate vegetation. I also identified that the workflow cannot yet prove additionality because it lacks a pre-project baseline, matched reference area, biomass calibration and field validation.

Likely interview questions

Question	Strong answer touchpoints
Why Surface Reflectance?	Atmospherically corrected; more suitable for temporal comparison than top-of-atmosphere reflectance.
Why not rely on scene cloud percentage?	It describes the full tile; project pixels still need Cloud Score+ and SCL-based masking.
Why a median composite?	Reduces residual clouds and extreme observations, but suppresses event timing and should not replace time-series analysis.
What was the strongest finding?	Seasonality dominates: NDVI falls during dry months and peaks during monsoon; this can mimic project impact.
What is the biggest limitation?	Single-year EDA cannot establish a baseline, causal additionality or claimable carbon.
What would you build next?	Reference-area matching, multi-year pre/post analysis, Sentinel-1 fusion, GEDI/field biomass model and uncertainty reporting.



19. Recommended 90-Day Implementation Roadmap

Period	Focus	Deliverables
Days 1-15	Requirements and data audit	Methodology, project boundary, start date, reference criteria, field-data inventory
Days 16-30	Multi-year baseline	Historical Landsat/Sentinel composites, quality metrics, pre-project vegetation map
Days 31-45	Change detection	Growth, canopy loss, cutting and regrowth event candidates
Days 46-60	Reference comparison	Matched reference area and difference-in-differences trend analysis
Days 61-75	Biomass and carbon	GEDI/field calibration, biomass estimates, uncertainty intervals
Days 76-90	Deployment and interview readiness	Dashboard, auditable report, APIs, monitoring, mock interview demonstration

Final project success means the system can explain what changed, where, when, how confidently, and which part may be attributable to the intervention - while preserving human verification.

End of report